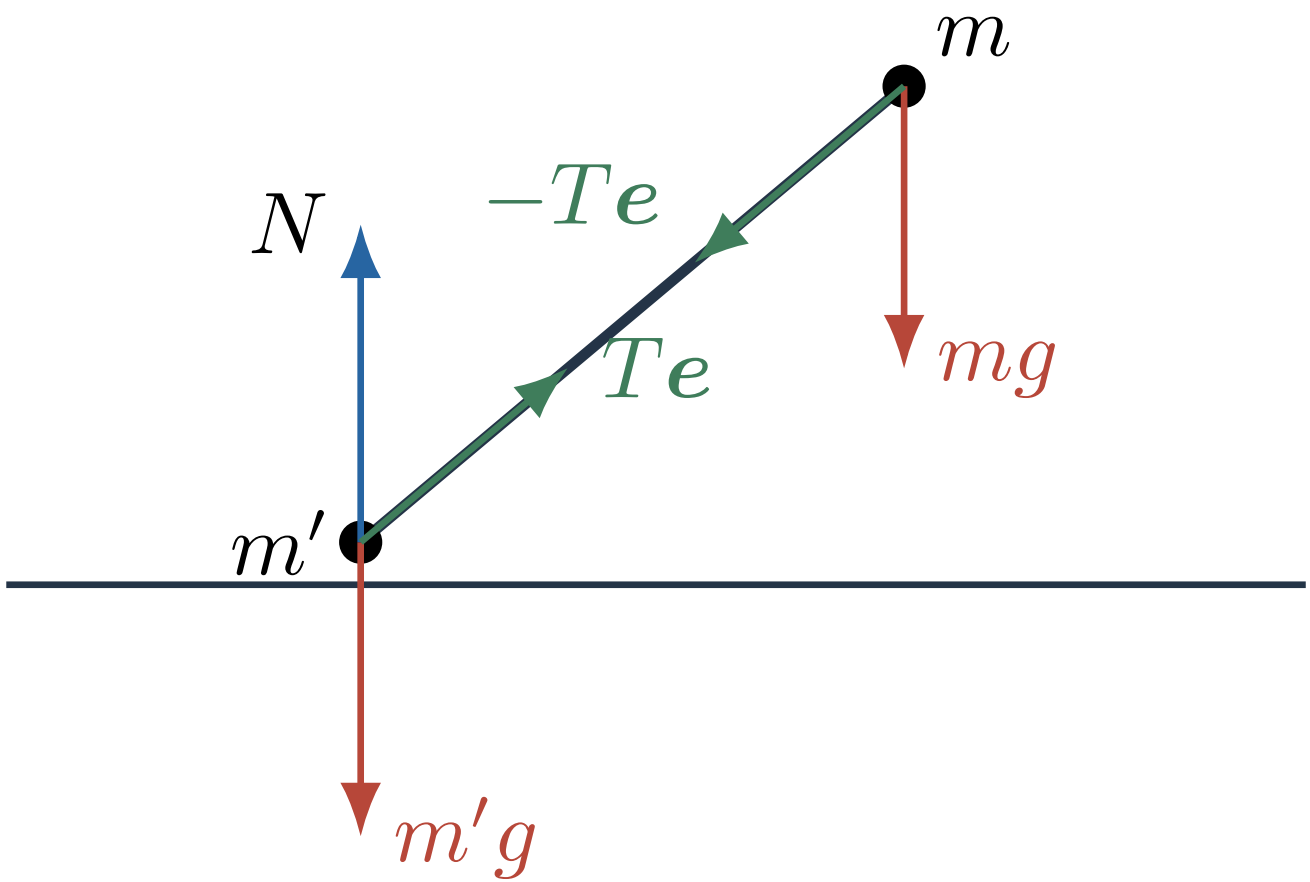

$T > 0$：拉力；$T < 0$：压力

无磁场时底端竖直平衡：$N = m'g - T\sin\theta$